

Problem 6.6

An amount of money is invested for one year at a rate of interest of 3% per quarter. Let $D(k)$ be the difference between the amount of interest earned on a compound interest basis, and on a simple interest basis for quarter k , where $k = 1, 2, 3, 4$. Find the ratio of $D(4)$ to $D(3)$.

Let the principal be P .

For quarter k , the simple interest earned is constant:

$$I_s(k) = 0.03P.$$

The compound interest earned during quarter k is the increase from the end of quarter $k - 1$ to the end of quarter k :

$$I_c(k) = P(1.03)^k - P(1.03)^{k-1} = 0.03P(1.03)^{k-1}.$$

Therefore,

$$D(k) = I_c(k) - I_s(k) = 0.03P((1.03)^{k-1} - 1).$$

Now compute $D(4)$ and $D(3)$:

$$D(4) = 0.03P((1.03)^3 - 1),$$

$$D(3) = 0.03P((1.03)^2 - 1).$$

Hence,

$$\frac{D(4)}{D(3)} = \frac{(1.03)^3 - 1}{(1.03)^2 - 1}.$$

Using

$$(1.03)^2 = 1.0609, \quad (1.03)^3 = 1.092727,$$

we get

$$\frac{D(4)}{D(3)} = \frac{1.092727 - 1}{1.0609 - 1} = \frac{0.092727}{0.0609} \approx 1.5226.$$

Therefore,

$$\boxed{\frac{D(4)}{D(3)} \approx 1.5226}.$$